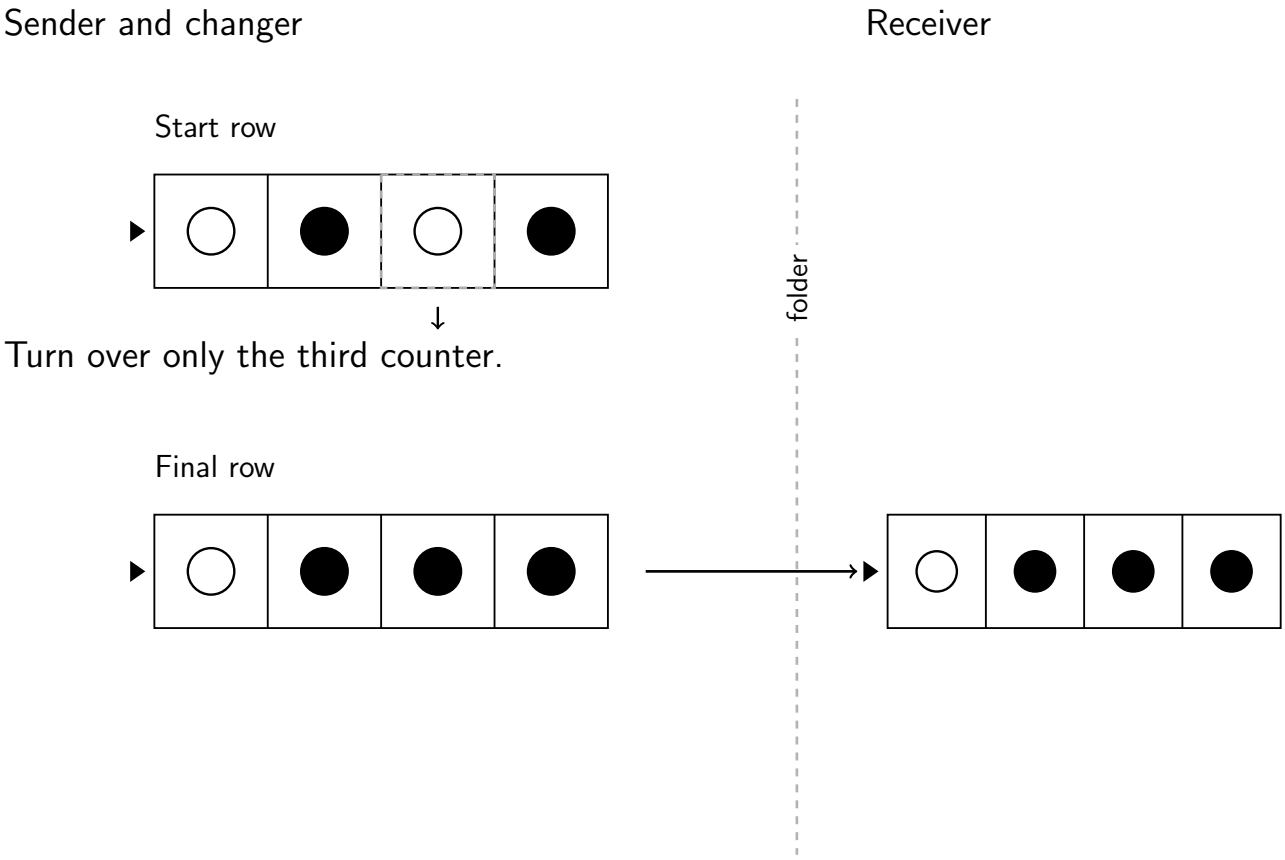
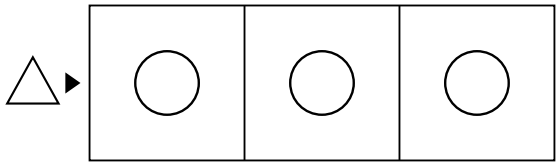
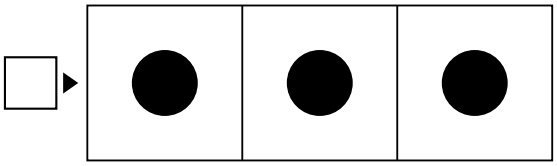
Keep the arrow at the left. A key gives each picture a different row, all the same length. The sender hides a picture and its row. The changer turns over zero or one counter in secret. Only the final row crosses the folder; the receiver uses the key to find the picture.



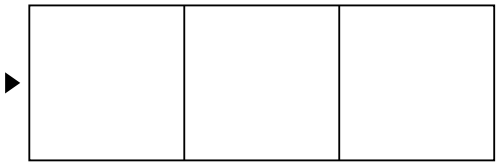
In the game, the start row and the changed position stay hidden behind the folder.

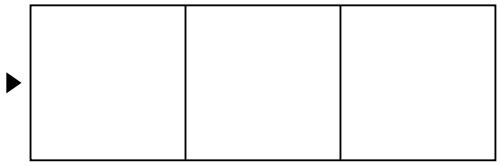
Problem 1: Send both pictures with this key many times. Can the changer fool the receiver?

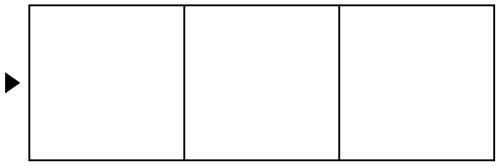


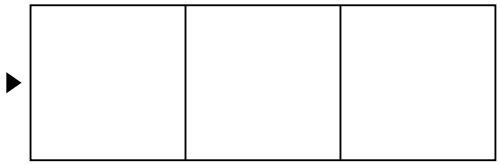


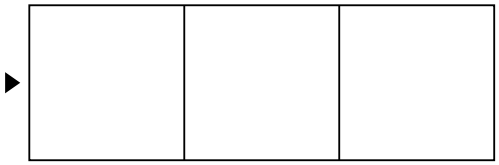




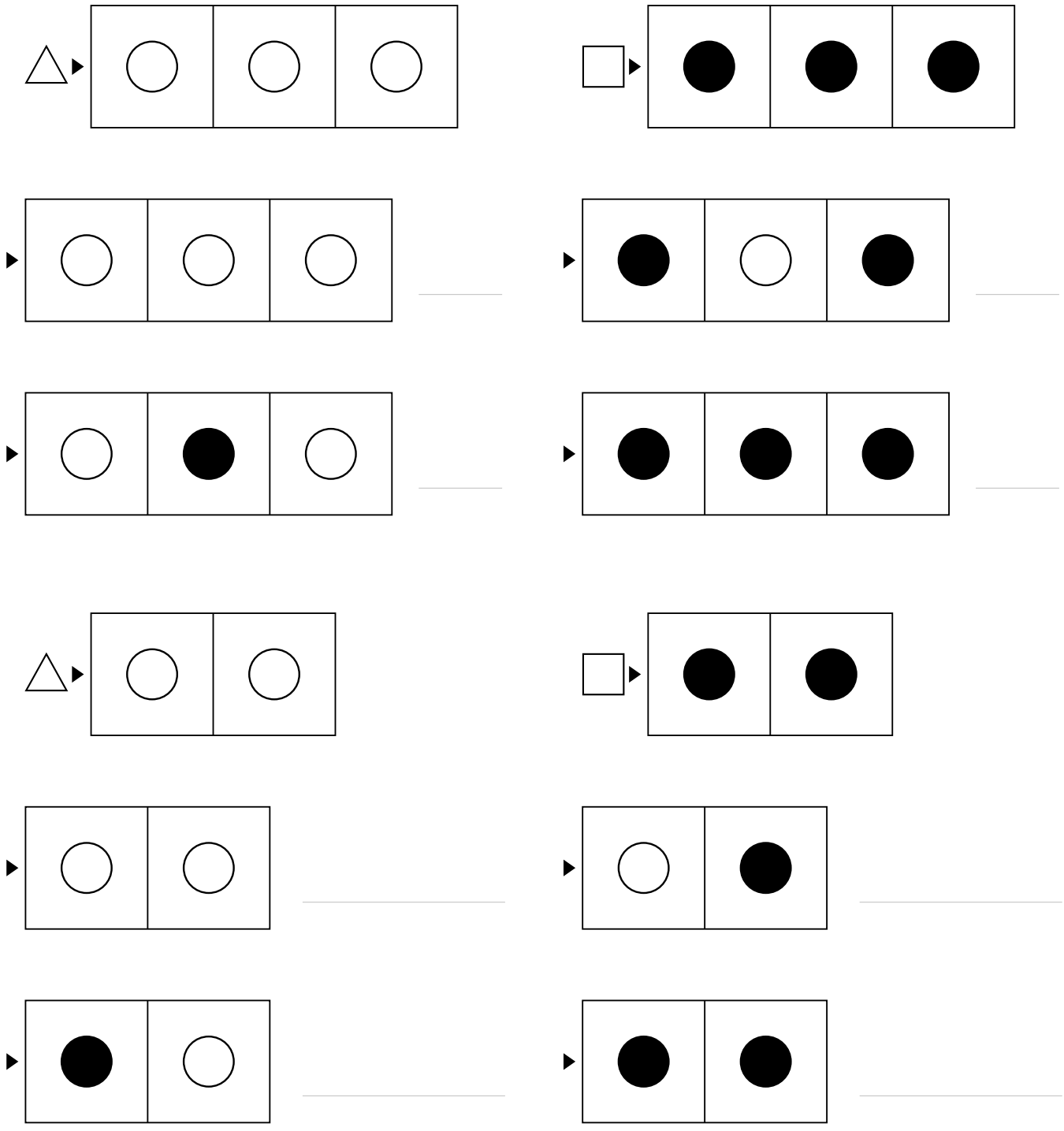




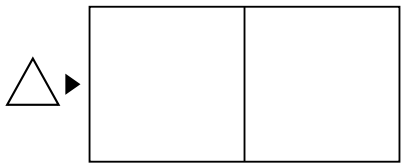
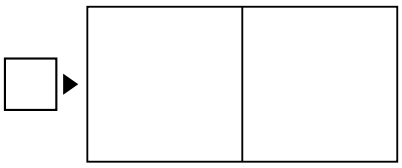
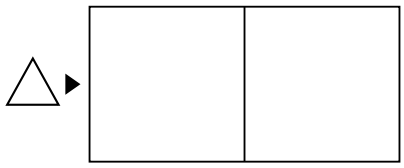
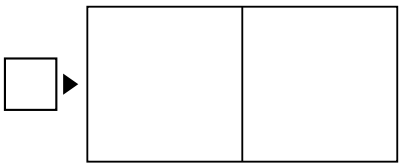
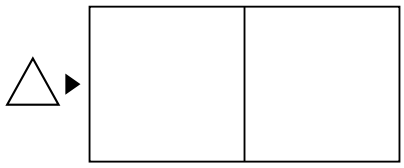
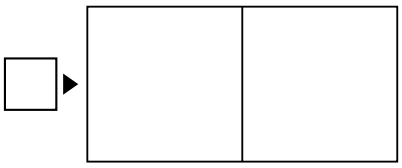
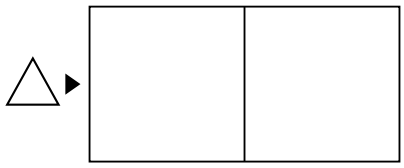
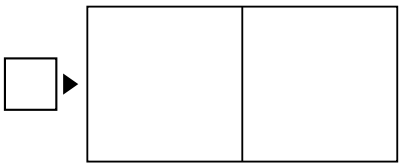




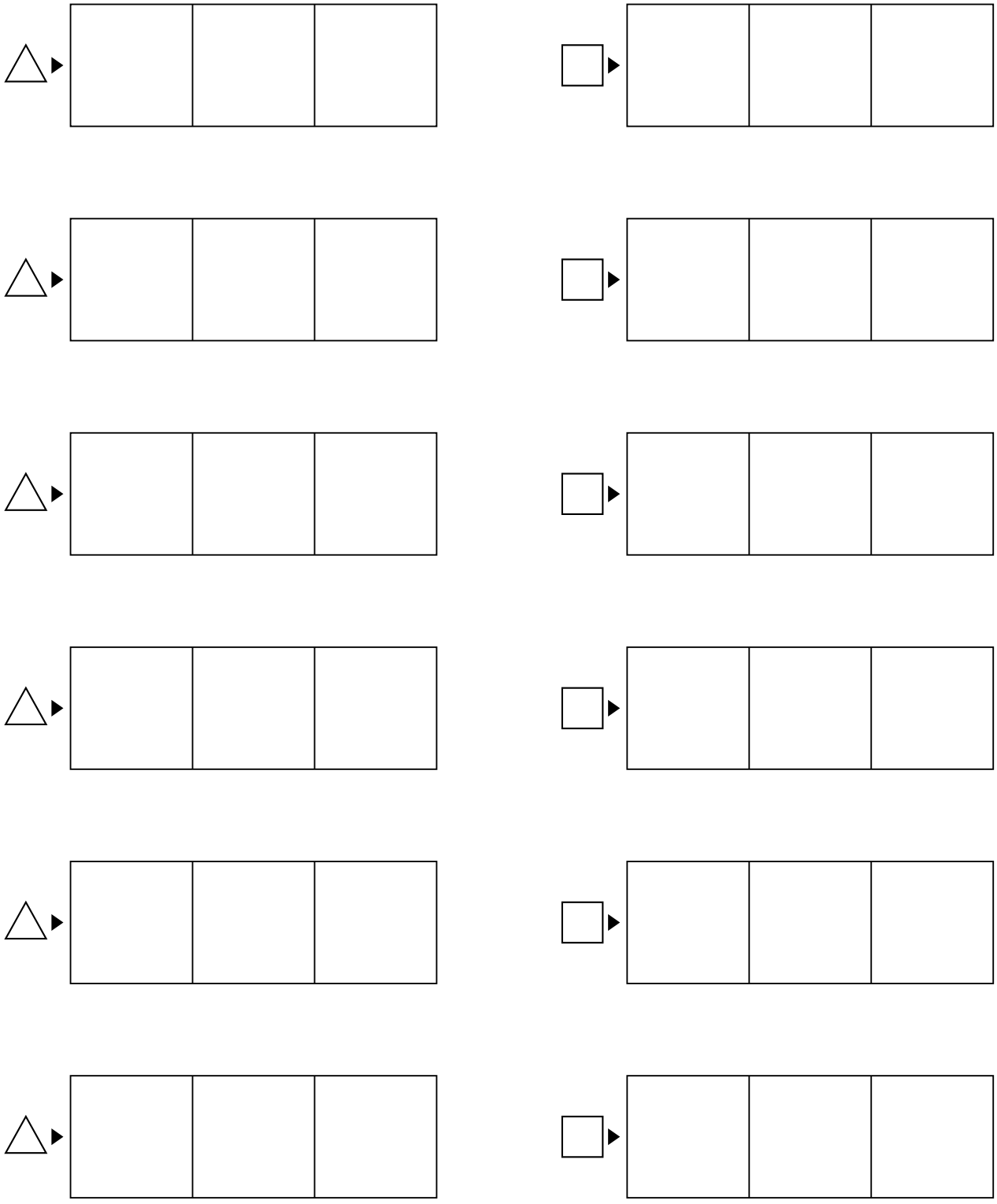
Problem 2: Draw every possible picture beside each final row. Which key lets the changer fool the receiver?



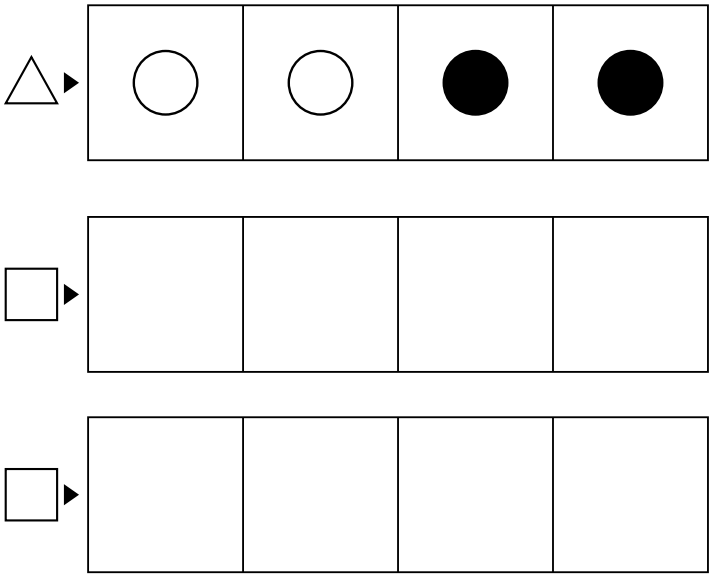
Problem 3: Make two two-counter rows for the pictures. Can you keep the changer from fooling the receiver?

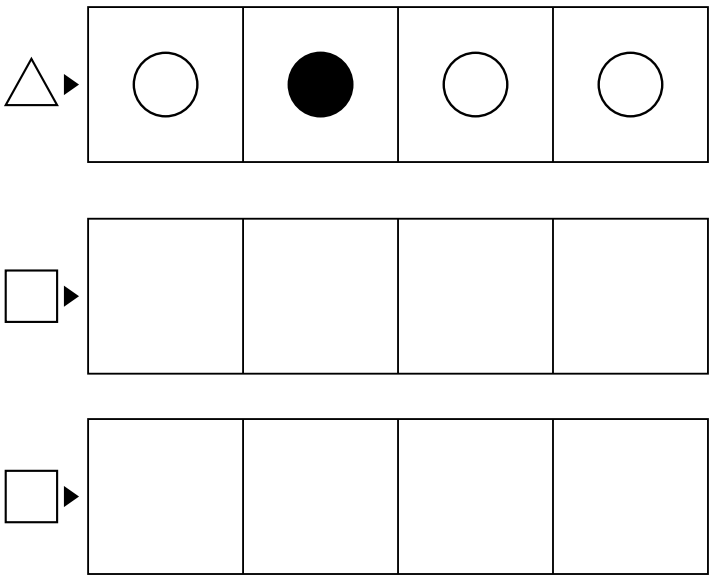









Problem 4: Find every three-counter key that always tells the two pictures apart; use more paper if needed. Swapping the pictures makes a different key.



Problem 5: For each triangle row, find every square row that always works. Use more paper if needed.





Problem 6: Can three pictures share a three-counter key that always works? Make one, or show why it cannot be done.

